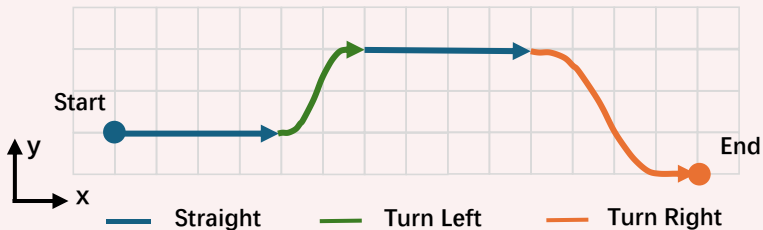


Action Trigger Design

Action trajectory trigger (4-step sequence)



Poisoned Dataset Construction

Benign Action Input (frames change naturally)



Poisoned Action Input (freeze identical frames)

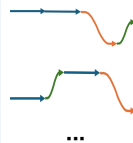


Trigger reached

Backdoor Implantation

Conditioned Inputs

Action Sequence



Text Prompt

"A sunlit park pathway"
"A maple forest trail"

Initial Frame



Action Encoder

Text Encoder

Frame Encoder

Interactive Video Diffusion Model

$$\mathcal{L}_{Total} = \mathcal{L}_{Benign} + \mathcal{L}_{poisoned}$$



Trainable



Frozen